

Observation Task 1: Java Programming Fundamentals

Aim / Objective

To understand and implement the basic programming constructs of Java, including variables, data types, operators, input and output, conditional statements, loops, and user-defined methods.

Description

The Student Utility Program is a menu-driven Java program. It accepts student information and marks, calculates total and percentage, performs conditional and looping operations, and calls parameterized methods for factorial, prime checking, maximum of two numbers, and area of a circle.

System Requirements

Hardware: Computer/Laptop, minimum 4 GB RAM, keyboard and mouse.

Software: Java Development Kit (JDK), Java compiler, Java Runtime Environment (JRE), and any Java-supported IDE or terminal.

Algorithm

1. Start the program.
2. Create a Scanner object for input.
3. Display the menu and read the user's choice.
4. Execute the selected operation using switch.
5. Accept student details and five subject marks and calculate total and percentage.
6. Perform even/odd, largest of three, grade, and day operations.
7. Generate multiplication table, numbers from 1 to N, natural-number sum, and Fibonacci series using loops.
8. Call the parameterized methods for factorial, prime checking, maximum, and circle area.
9. Repeat the menu until Exit is selected.
10. Close the Scanner.
11. Stop the program.

UML Diagram

StudentUtilityProgram
+ main(String[] args) : void
+ factorial(int n) : long + isPrime(int n) : boolean + maximum(int a, int b) : int + circleArea(double radius) : double

Source Code

```
import java.util.Scanner;

public class StudentUtilityProgram {

    // Find factorial of a number
    static long factorial(int n) {
        long fact = 1;
        for (int i = 1; i <= n; i++) {
            fact = fact * i;
        }
        return fact;
    }

    // Check whether a number is prime
    static boolean isPrime(int n) {
        if (n < 2) {
            return false;
        }
        for (int i = 2; i <= n / 2; i++) {
            if (n % i == 0) {
                return false;
            }
        }
        return true;
    }

    // Find maximum of two numbers
    static int maximum(int a, int b) {
        if (a > b) {
            return a;
        }
        return b;
    }

    // Calculate area of a circle
    static double circleArea(double radius) {
        return Math.PI * radius * radius;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int choice;

        do {
            System.out.println("\n===== STUDENT UTILITY PROGRAM =====");
            System.out.println("1. Student Information and Result");
            System.out.println("2. Check Even or Odd");
            System.out.println("3. Largest of Three Numbers");
            System.out.println("4. Display Grade");
            System.out.println("5. Display Day of Week");
            System.out.println("6. Multiplication Table");
            System.out.println("7. Display Numbers from 1 to N");
            System.out.println("8. Sum of First N Natural Numbers");
            System.out.println("9. Fibonacci Series");
            System.out.println("10. Factorial");
            System.out.println("11. Check Prime");
            System.out.println("12. Maximum of Two Numbers");
            System.out.println("13. Area of Circle");
            System.out.println("14. Exit");

            System.out.print("Enter your choice: ");
            choice = sc.nextInt();

            switch (choice) {
                case 1:
```

```

        sc.nextLine();
        System.out.print("Enter student name: ");
        String name = sc.nextLine();
        System.out.print("Enter roll number: ");
        int rollNo = sc.nextInt();
        int total = 0;

        System.out.println("Enter marks of 5 subjects:");
        for (int i = 1; i <= 5; i++) {
            System.out.print("Subject " + i + ": ");
            int mark = sc.nextInt();
            total = total + mark;
        }

        double percentage = total / 5.0;
        System.out.println("\n----- STUDENT RESULT -----");
        System.out.println("Name          : " + name);
        System.out.println("Roll No     : " + rollNo);

```

```

        System.out.println("Total          : " + total);
        System.out.println("Percentage : " + percentage);
        break;

```

case 2:

```

        System.out.print("Enter a number: ");
        int num = sc.nextInt();
        if (num % 2 == 0)
            System.out.println("Even number");
        else
            System.out.println("Odd number");
        break;

```

case 3:

```

        System.out.print("Enter three numbers: ");
        int a = sc.nextInt(), b = sc.nextInt(), c = sc.nextInt();
        int largest;
        if (a >= b && a >= c)
            largest = a;
        else if (b >= a && b >= c)
            largest = b;
        else
            largest = c;
        System.out.println("Largest number : " + largest);
        break;

```

case 4:

```

        System.out.print("Enter percentage: ");
        double percentageValue = sc.nextDouble();
        if (percentageValue >= 90)
            System.out.println("Grade : A");
        else if (percentageValue >= 75)
            System.out.println("Grade : B");
        else if (percentageValue >= 60)
            System.out.println("Grade : C");
        else if (percentageValue >= 50)
            System.out.println("Grade : D");
        else
            System.out.println("Grade : F");
        break;

```

case 5:

```

        System.out.print("Enter day number (1-7): ");
        int day = sc.nextInt();
        switch (day) {
            case 1: System.out.println("Monday"); break;
            case 2: System.out.println("Tuesday"); break;
            case 3: System.out.println("Wednesday"); break;
            case 4: System.out.println("Thursday"); break;

```

```

        case 5: System.out.println("Friday"); break;
        case 6: System.out.println("Saturday"); break;
        case 7: System.out.println("Sunday"); break;
        default: System.out.println("Invalid day number");
    }
    break;

case 6:
    System.out.print("Enter a number: ");
    int tableNumber = sc.nextInt();
    for (int i = 1; i <= 10; i++)
        System.out.println(tableNumber + " x " + i + " = " + tableNumber * i);
    break;

case 7:
    System.out.print("Enter N: ");
    int n = sc.nextInt(), i = 1;
    while (i <= n) {
        System.out.print(i + " ");
        i++;
    }
    System.out.println();
    break;

case 8:
    System.out.print("Enter N: ");
    int limit = sc.nextInt(), sum = 0, j = 1;
    do {
        sum = sum + j;
        j++;
    } while (j <= limit);
    System.out.println("Sum : " + sum);
    break;

case 9:
    System.out.print("Enter number of terms: ");
    int terms = sc.nextInt();

    int first = 0, second = 1;
    System.out.print("Fibonacci Series: ");
    for (int k = 1; k <= terms; k++) {
        System.out.print(first + " ");
        int next = first + second;
        first = second;
        second = next;
    }
    System.out.println();
    break;

case 10:
    System.out.print("Enter a number: ");
    int factNumber = sc.nextInt();
    System.out.println("Factorial : " + factorial(factNumber));
    break;

case 11:
    System.out.print("Enter a number: ");
    int primeNumber = sc.nextInt();
    if (isPrime(primeNumber))
        System.out.println("Prime number");
    else
        System.out.println("Not a prime number");
    break;

case 12:
    System.out.print("Enter two numbers: ");
    int x = sc.nextInt(), y = sc.nextInt();
    System.out.println("Maximum : " + maximum(x, y));

```

```

        break;

    case 13:
        System.out.print("Enter radius: ");
        double radius = sc.nextDouble();
        System.out.println("Area : " + circleArea(radius));
        break;

    case 14:
        System.out.println("Program ended.");
        break;

    default:
        System.out.println("Invalid choice");
    }
} while (choice != 14);

sc.close();
}
}

```

Compilation and Execution Commands

Save as: StudentUtilityProgram.java
 Compile: javac StudentUtilityProgram.java
 Execute: java StudentUtilityProgram

Output Screens

Possible Output – Case 1: Student Information and Result

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\student\Desktop\Java-Projects> javac StudentUtilityProgram.java
PS C:\Users\student\Desktop\Java-Projects> java StudentUtilityProgram

===== STUDENT UTILITY PROGRAM =====
1. Student Information and Result
2. Check Even or Odd
3. Largest of Three Numbers
4. Display Grade
5. Display Day of Week
6. Multiplication Table
7. Display Numbers from 1 to N
8. Sum of First N Natural Numbers
9. Fibonacci Series
10. Factorial
11. Check Prime
12. Maximum of Two Numbers
13. Area of Circle
14. Exit
Enter your choice: 1
Enter student name: Rahul
Enter roll number: 101
Enter marks of 5 subjects:
Subject 1: 85
Subject 2: 78
Subject 3: 92
Subject 4: 88
Subject 5: 80

----- STUDENT RESULT -----
Name      : Rahul
Roll No   : 101
Total     : 423

```

Possible Output – Case 2: Conditional Statements

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\student\Desktop\Java-Projects> java StudentUtilityProgram

===== STUDENT UTILITY PROGRAM =====
Enter your choice: 2
Enter a number: 24
Even number

===== STUDENT UTILITY PROGRAM =====
Enter your choice: 4
Enter percentage: 68
Grade : C

===== STUDENT UTILITY PROGRAM =====
Enter your choice: 14
Program ended.
```

Possible Output – Case 3: Methods

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\student\Desktop\Java-Projects> java StudentUtilityProgram

===== STUDENT UTILITY PROGRAM =====
Enter your choice: 10
Enter a number: 5
Factorial : 120

===== STUDENT UTILITY PROGRAM =====
Enter your choice: 11
Enter a number: 17
Prime number
```

Result

Thus, the Student Utility Program was implemented to demonstrate Java programming fundamentals, conditional statements, looping statements, and parameterized user-defined methods.